

WAVES QUESTIONS

1. (a) Briefly explain three applications of diffraction of waves.
(b) One scientist was investigating the impact of tension on the frequency of vibrating wire. The following data were recorded:
Length of a stretched wire was 30 cm, tension of wire was 900 N and slotted masses of 3 g and 12 g.
Use these data to show how the masses on the vibrating wire affects its frequency.
2. (a) A person strikes one end of a long iron pipe with a hammer. A student standing at the other end hears two distinct sounds separated by a time interval of 1.2 seconds. If the length of the iron pipe is 442 meters and the speed of sound in air is 340 m/s, calculate the speed of sound in the metal.
(b) A man is standing at point P between the walls X and Y shown in Figure 6 then claps his hands once. If the first two echoes are separated by 0.4 s, find the separation distance of the wall X and Y. (speed of sound in air = 330 m/s)

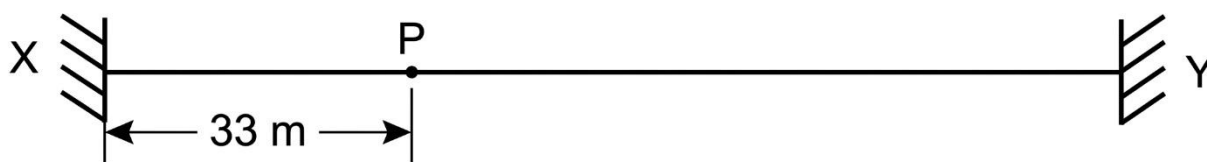


Figure 6

3. (a) A piano wire and tuning fork are devices which produce sound. Show clearly three ways in which a piano wire can be tuned to emit a note of the same frequency as a vibrating tuning fork.
(b) Figure 3 shows the profile of a radio wave. Study it carefully and determine its wavelength.

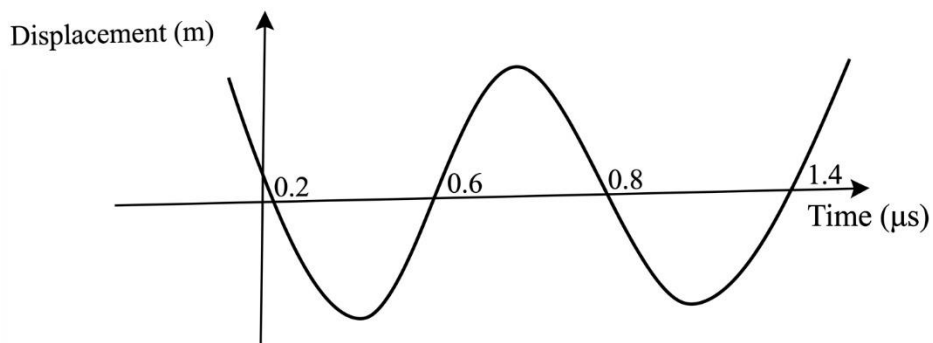


Figure 3

4. (a) A man standing 504 m from a cliff claps his hands and hears the echo 3 s later. What value does this observation give for the velocity of sound in air?
 (b) The frequency obtained from a plucked string when the tension is 2 N is 400 Hz. Calculate the frequency when the tension is increased by 6 N.
5. (a) A ship using an echo-sounding device receives an echo from a wreck 0.8 s after the sound is transmitted. If the velocity of sound in seawater is 1500 m/s, what is the depth of the wreck?
 (b) A resonance tube whose one end is closed and the other open, resonates to a note of frequency 560 Hz when the length of the air column is 15 cm. Determine the wavelength of this sound in air. What is the shortest length of the air column which resonates in similar conditions to a note of frequency 1000 Hz.
6. (a) Briefly explain how beats are formed.
 (b) A vibrating string is sounded with a 288 Hz tuning fork and 3 beats are counted per second. The string is then loaded with small amount of plasticine and 2 beats are counted per second.
 (i) What is the frequency of the string?
 (ii) What is the frequency of the loaded string?
7. (a) If the shortest length of a tube for resonance is 0.12 m and the next resonant length is 0.37 m, what is the frequency of vibration? Take the speed of sound in air as 340 m s^{-1} .
 (b) A man standing 510 m from a cliff fires a rifle. Another man standing nearer to the cliff hears two shots, the interval between their arrival being 1.0 s. What is the distance between the two men if the line joining them is at right angles to the cliff?
8. (a) A tuning fork produces resonances in a tube at 40 cm and 85 cm respectively. Calculate the wavelength of the tuning fork.
 (b) A stretched taut string of length 40 cm and mass $2 \times 10^{-3} \text{ kg}$ is pulled with a tension of 100 N. Determine the frequency of the first and second harmonics when it is plucked at its middle.
9. (a) Guitars have strings of varying thickness. Which of the strings (thickest or thinnest) produces the highest frequency of musical notes? Explain your answer.
 (b) With the aid of diagrams, show how the fundamental frequency and frequency of the first overtone can be obtained for a closed pipe.
 (c) Explain two ways how reverberations in a big hall or auditorium can be reduced.
10. (a) Why an echo does not occur in a small room?

- (b) Given that the velocity of the sound wave emitted from the string is 50 m/s. The length of the string 40 cm, and the mass of the string is 0.0004 kg. Calculate the tension of the string.
- (c) Briefly explain why during thunderstorm the sound is heard later after the flash was observed.
11. A stretched wire vibrates at 100 Hz. What is the frequency of a wire of the same material and thickness which is
- twice as long,
 - the same length, but with four times the tension?
12. (a) A horizontal wire 2.0 m long gives a note of frequency 150 Hz. A 'bridge' is placed under the wire 0.50 m from one end. What are the frequencies produced by the two segments of the wire?
- (b) A string is set vibrating by plucking at a point one-quarter of the length from the end.
- Draw a sketched diagram showing nodes and antinodes in a vibrating string. (03 marks)
 - Write the expression of the frequency note produced by the vibrating string. (03 marks)
13. (a) A sonometer wire between a fixed bridge X and an adjustable bridge Y has a frequency of 320 Hz when Y is 0.80 m from X. How far must Y be moved to produce a frequency of 256 Hz? Which way must Y be moved?
- (b) Two men stand a distance apart beside a long metal fence on a still day. One man places his ear against the fence, while the other gives the fences a sharp knock with a hammer two sound separated by a time interval of 0.5 seconds are heard by the first man. If the velocity of sound in air is 330m/s and in the metal is 5280m/s. How far apart are the men?
14. (a) Find the frequency of the fundamental note given by a stretched wire if it is 0.60 m long, has a mass of 10 g and a tension of 960 N.
- (b) Figure 1.58 shows the main bands of the electromagnetic spectrum. Region D represents visible light.

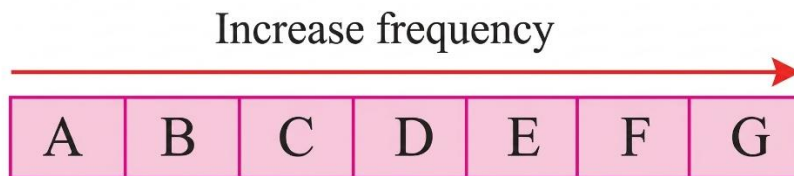


Figure 1.58

- (i) Which region represents radiation capable of promoting the production of vitamin D in the skin?
 - (ii) Which region contains radiation used in RADAR systems?
 - (iii) Which region contains radiation produced in nuclear reactors?
15. (a) A stretched string is in unison with a tuning fork of frequency 384 Hz. If the tension in the string is increased by 2%, what is the frequency of the beats heard when the fork and the string are now sounded together?
- (b) Figure 2 and 3 shows a wave travelling across water. Carefully study it then answer the questions that follow:

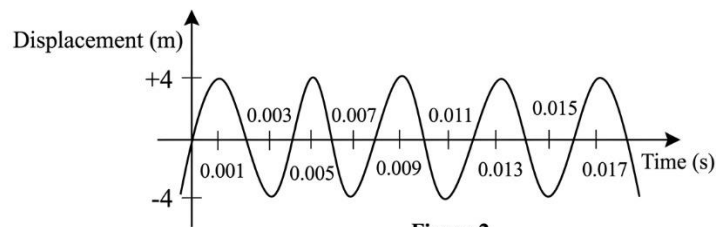


Figure 2

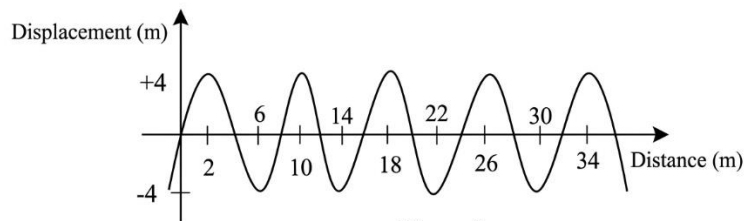


Figure 3

- (i) Determine the frequency of the wave
 - (ii) Determine the velocity of the wave
16. Water waves of amplitude 12 mm travel at 16 cm/s. Their wavelength is 20 mm.
- (a) What is their frequency?
 - (b) Draw a full-size diagram to show a side-view of two complete waves. Label the amplitude and wavelength.
17. (a) If light travels at 3.00×10^8 m/s in a vacuum what is its speed in glass of refractive index 1.5?

(b) Light of wavelength 6.0×10^{-7} m in a vacuum enters the glass. What is its wavelength in the glass?

18. Fig. 43 shows a displacement-time graph for a wave on the surface of a liquid. What is
- (a) the amplitude of the wave?
 - (b) the period?
 - (c) the frequency?
 - (d) The wavelength is 18 mm. What is the velocity?

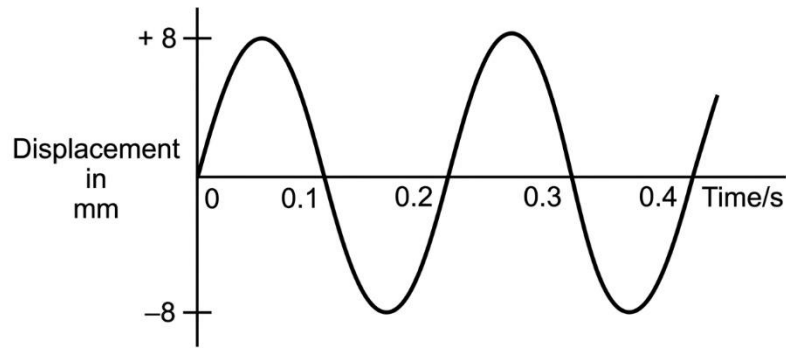


Fig. 43

19. Two persons stand facing each other, 200 m apart on one side of a high wall and at the same perpendicular distance to it. When one fires a pistol, the other hears a report 0.60 s after the flash and a second 0.25 s after the first. Explain this and calculate:
- (a) Draw a diagram showing the positions of the persons and the wall
 - (b) The velocity of sound in air.
 - (c) The perpendicular distance of the persons from the wall.
20. (a) Calculate the ratio of velocities of sound produced in dry air at 42°C to another at 60°C .
- (b) (i) Briefly, explain how resonance tube works?
 - (ii) Calculate the frequency of vibration in resonance tube of shortest length of 0.22 m when next resonance length is 0.47 m.
21. (a) How long would it take for a 30 Hz beat to reach an audience member 100 m away when the ambient temperature is 21°C ?
- (b) A sound is sent out from the ship and its reflection from the ocean floor returns after a second later. Assuming that the velocity of sound in water is 1500 m/s, how deep is the ocean?
22. An observer stands between two distant cliffs and claps hands. An echo is received after 2 s and 2.5 s respectively. If the speed of sound in air is 330 m s⁻¹, find the distance between the cliffs.

23. (a) Explain why radio waves are similar to light waves but not sound waves
- (b) In a closed pipe, the first resonance is at 23 cm and second at 73 cm. determines the wave length of the sound and the end correction of pipe
- (c) Matter expands when heated and contracts when cooled. Explain why a musician must retune a stringed instrument if its temperature changes
24. (a) If you want to hit a fish under water using a spear, should you aim below it, straight at it or above it? Explain.
- (b) How speed and wavelength are affected when wave move from one medium to another.
25. (a) A periodic disturbance in a lake creates waves which emanate outwards from its source to produce circular wave patterns. If the frequency of the source is 2 Hz and the wave speed is 5 m/s, determine the distance between adjacent wave crests.
- (b) A girl claps her hands at distance of 250m from a vertical cliff. How long does it take for her to hear her echo? If the speed of sound was 340m/s.
26. A wave whose speed in the first medium is 4.4 m/s enters a second medium. The wavelength changes from 2 m to 3 m. What is the speed of the wave in the second medium?
27. A sinusoidal wave is travelling along a rope. The oscillator that generates the wave completes 40 vibrations in 30 s. Also, a given crest of the wave travels 425 cm along the rope in 10 s. Calculate the wavelength of the sinusoidal wave.
28. The velocity of light in water is $2.2 \times 10^8 \text{ ms}^{-1}$ and the velocity of light in glass is $2.0 \times 10^8 \text{ ms}^{-1}$. Calculate:
- (a) The relative refractive index as light passes from water to glass.
- (b) The angle of incidence in the water which would produce an angle of refraction of 30° in the glass.
29. A straight vibrator causes water ripples to travel across the surface of a shallow tank. The ripples travel a distance of 33 cm in 1.5 s and the distance between successive crests of the wave is 4.0 cm. Calculate the frequency of the vibrator.
30. Radio and light waves travel at a velocity of $3 \times 10^8 \text{ m/s}$ in air. Calculate:
- (a) The wavelength of radio waves when transmitted at a frequency 150 MHz; and
- (b) The velocity of light in glass of refractive index 1.5.

31. (a) At a beach, a lifeguard observes waves approaching the shore. He notices that the wave passes a buoy three times every second, and the distance between the two successive crests is 2m. How fast is the wave moving?
- (b) Why do recording studios and concert auditoriums sometimes have their ceilings and walls covered with soft perforated boards?
32. A string of length 1.2 m is stretched and made to vibrate so that a stationary wave consisting of two loops is produced.
- (a) Draw a sketch of the wave.
- (b) Determine the wavelength of the wave.
33. wire of length 20 cm, mass 1.2 g and under a tension of 120 N is plucked to generate a wave. Determine:
- (a) The fundamental frequency.
- (b) The frequency of the third harmonic.
34. A speaker delivering a note of frequency 250 Hz is placed over the upper end of a vertical tube filled with water. When the water is gradually run down the tube, the air column resonates when the water level is 31 cm below the top of the tube. The air column resonates again when the water level is 99.8 cm below the top of the tube. Determine:
- (a) The speed of sound in air.
- (b) The end correction.
35. If the shortest length of a tube for resonance is 0.12 m and the next resonant length is 0.37 m, what is the frequency of vibration? Take the speed of sound in air as 340 m/s.
36. A column of air 26.25 cm long in a closed tube resonates to a sounding tuning fork. If the velocity of sound in air is 33600 cm s^{-1} , what is the frequency of the fork? Ignore end correction.
37. During a storm, thunder is heard 7 s after the lightning is seen. If the temperature of the air at the time of the storm is 28°C , how far away is the storm cloud?
38. A loud sound is made and an echo from a distant cliff is heard 8 s later. If the atmospheric temperature is 22°C , how far away is the cliff?
39. String A is 2 m long and has a linear mass density of 9 g cm^{-1} . String B has a linear mass density of 18 g cm^{-1} . If the tension in both strings is the same, how long must string B be for it to be at resonance with string A?

40. In a resonance tube experiment, the smallest value of l for which a peak in sound intensity occurs is 9.0 cm. How far must the tube be raised to hear the next peak in sound intensity? Neglect the end correction.
41. Explain why it is not advisable for soldiers to march across a bridge in rhythm.
42. A sonometer wire was tuned to a fundamental frequency of 250 Hz. The vibrating wire was then under tension of 10 N and was 87 cm long. Show by calculation how to tune the wire to a fundamental frequency of 375 Hz by adjusting:
- (a) Its length only.
 - (b) Tension only.
43. Consider two organ pipes A and B sounded with their:
- (i) Fundamental notes, and
 - (ii) First overtones. If pipe A is closed at one end and pipe B is open at both ends, draw diagrams to show the mode of vibration in each of the four cases.
44. (a) How does the size of a gap in a barrier affect the diffraction of waves?
- (b) (i) State two ways in which visible light differs from radio waves.
 - (ii) List two applications of gamma rays.
45. (a) (a) (i) List four main parts of a ripple tank.
- (ii) What role does a stroboscope play in a ripple tank experiment?
 - (b) (i) Explain why there are four strings of different thickness in a violin?
 - (ii) What does a violinist do to change the note emitted by a particular string?
46. Explain why
- (a) notes of the same pitch played on a violin and a flute sound different,
 - (b) the lowest pitched note played on a flute is lower than that played on a piccolo (a similar but smaller wind instrument).

SOLUTIONS

1. (a) Applications of diffraction of waves

- (i) Diffraction is used in a hologram to generate a three-dimensional impression of an image.
- (ii) Diffraction is used in measuring the coefficient of thermal expansion, crystallite size and thickness of thin films.
- (iii) X-rays diffraction is used to determine the distance between two consecutive atoms of a material.

(b) Required to show how the masses on the vibrating wire affects its frequency

It has been shown that

$$f \propto \frac{1}{\sqrt{\mu}}, \text{ where } \mu = \frac{\text{mass}}{\text{length}}$$

$$f\sqrt{\mu} = \text{constant}$$

$$f_1 \times \sqrt{\mu_1} = f_2 \times \sqrt{\mu_2}$$

since length of a string is constant

$$f_1 \times \sqrt{m_1} = f_2 \times \sqrt{m_2}$$

$$f_1 \times \sqrt{3g} = f_2 \times \sqrt{12g}$$

$$f_1 \times \sqrt{\frac{3g}{12g}} = f_2$$

$$f_1 \times \frac{1}{2} = f_2$$

Therefore, frequency of vibration decreases when the mass increases.

2. (a) Required to calculate the speed of sound in the metal

From the formula

$$\text{Speed of sound (v)} = \frac{\text{Distance covered (d)}}{\text{Time taken (t)}}$$

$$t = \frac{d}{v}$$

When sound travels through air when

$$t_1 = d/v_a$$

When sound travels through the metal

$$t_2 = d/v_m$$

Since sound travels faster in metal (solid) than air (gas)

$$t_1 - t_2 = \Delta t = d/v_a - d/v_m = d \times (1/v_a - 1/v_m)$$

$$\frac{\Delta t}{d} = 1/v_a - 1/v_m$$

$$1/v_m = 1/v_a - \Delta t/d = (d - v_a \Delta t)/(v_a \times d)$$

$$v_m = (v_a \times d)/(d - v_a \Delta t) = (340 \text{ m/s} \times 442 \text{ m})/(442 \text{ m} - 340 \text{ m/s} \times 1.2 \text{ s}) = 4420 \text{ m/s}$$

Therefore, the speed of sound in metal is 4420 m/s.

(b) Required to find the separation distance between the wall X and Y

From

$$\text{Speed of sound} = \frac{2 \times \text{Distance covered}}{\text{Time taken}}$$

$$\text{Time taken (t)} = \frac{2 \times \text{Distance covered}}{\text{speed of sound}}$$

The first echo is caused by the near wall X

$t_1 = 2d_1/v$, where d_1 is the distance between wall X and the man and v is the speed of sound in air

The second echo is caused by the far wall Y

$t_2 = 2d_2/v$, where d_2 is the distance between the man and wall Y

$$\Delta t = t_2 - t_1 = 2d_2/v - 2d_1/v = \frac{2}{v} \times (d_2 - d_1)$$

$$\frac{v \Delta t}{2} = d_2 - d_1$$

$$d_2 = \frac{v \Delta t}{2} + d_1 = \frac{330 \frac{\text{m}}{\text{s}} \times 0.4 \text{ s}}{2} + 33 \text{ m} = 99 \text{ m}$$

The distance between two walls (d) = $d_1 + d_2 = 33 \text{ m} + 99 \text{ m} = 132 \text{ m}$.

Therefore, the distance between the two walls is 132 m.

3. (a) The following below are the ways on how a piano wire can be turned to emit a note of the same frequency as a vibrating fork.

- (i) **Length of a piano wire:** The frequency produced by piano is affected by length of the wire, the shorter the wire, the higher the frequency produced.

$$f \propto \frac{1}{l}$$

- (ii) **Tension of the wire:** The frequency produced by piano is also affected by the tension of the piano wire. This means that the taut the wire, the higher the tension.

$$f \propto \sqrt{T}$$

- (iii) **Linear mass density:** Linear mass density of the wire also affects the frequency of the piano. This means that thin wire produces higher frequency while thick wire produces lower frequency.

$$f \propto \frac{1}{\sqrt{\mu}}$$

- (b) Required to determine the wavelength of wave.

From wave equation

$$v = f\lambda$$

$$\lambda = \frac{v}{f} = vT$$

$$T = 0.8 \mu\text{s} - 0.2 \mu\text{s} = 0.6 \mu\text{s} = 6 \times 10^{-7} \text{ s}$$

Since, radio wave is an electromagnetic wave, $v = 3.0 \times 10^8 \text{ m/s}$

$$\lambda = 3.0 \times 10^8 \text{ m/s} \times 6 \times 10^{-7} \text{ s} = 180 \text{ m.}$$

Therefore, the wavelength of radio wave is 180 m.

4. (a) Required to determine the velocity of sound in air

From

$$v = \frac{2d}{t}$$

$$v = \frac{2 \times 504 \text{ m}}{3 \text{ s}} = 336 \text{ m/s}$$

Therefore, the velocity of sound in air is 336 m/s.

- (b) Required to calculate the frequency a plucked string

From

$$f = k\sqrt{T}$$

$$f_1 = k\sqrt{T_1}$$

$$f_2 = k\sqrt{T_2}$$

$$f_2/f_1 = \sqrt{\left(\frac{T_2}{T_1}\right)}$$

$$f_2 = f_1 \times \sqrt{\left(\frac{T_2}{T_1}\right)}, \text{ where } f_1 = 400 \text{ Hz, } T_1 = 2 \text{ N and } T_2 = T_1 + \Delta T = 2 \text{ N} + 6 \text{ N} = 8 \text{ N}$$

$$f_2 = 400 \text{ Hz} \times \sqrt{\left(\frac{8\text{N}}{2\text{N}}\right)} = 800 \text{ Hz.}$$

Therefore, the new frequency is 800 Hz.

5. (a) Required to find the depth of the wreck

From

$$v = \frac{2d}{t}, \text{ where } d \text{ is the depth, } v \text{ is the speed of sound in water and } t \text{ is the time taken}$$

$$d = \frac{vt}{2} = \frac{1500 \frac{\text{m}}{\text{s}} \times 0.8 \text{ s}}{2}$$

$$d = 600 \text{ m}$$

Therefore, the depth of the wreck is 600 m.

- (b) Wavelength of the sound in air, $\lambda = ?$

From

$$l = \frac{\lambda}{4}$$

$$\lambda = 4l = 4 \times 15 \text{ cm} = 60 \text{ cm} = 0.6 \text{ m}$$

The wavelength of sound in air is 0.6 m.

The shortest length, $l = ?$ When $f = 1000 \text{ Hz}$

Initially

$$v = f\lambda \text{ (wave equation)}$$

$$\text{Where, } f = 560 \text{ Hz and } \lambda = 0.6 \text{ m}$$

$$v = 560/\text{s} \times 0.6 \text{ m} = 336 \text{ m/s}$$

Now

$$\lambda = \frac{v}{f} = \frac{336 \text{ m/s}}{1000/\text{s}} = 0.336 \text{ m.}$$

$$l = \frac{\lambda}{4} = \frac{0.336 \text{ m}}{4} = 0.084 \text{ m} = 8.4 \text{ cm}$$

Therefore, the shortest length of the air column is 8.4 cm.

6. (a) Beats are formed when two sounds of nearly equal frequencies are sounded together

(b) Data given

Frequency of vibrating string = f_1

Frequency of tuning fork, $f_2 = 288 \text{ Hz}$.

Beat frequency when tuning fork is not loaded, $f_b = 3 \text{ Hz}$

Beat frequency when string is loaded, $f_b = 2 \text{ Hz}$

From

Beat frequency (f_b) = $f_1 - f_2$ because when a string is loaded with plasticine, the frequency decreases. So $f_1 - f_2$ will give us small beat frequency when a string is loaded while $f_2 - f_1$ will give us higher beat frequency.

(i) Required to find the frequency of the string (f_1)

When tuning fork not loaded

$$f_1 - 288 \text{ Hz} = 3 \text{ Hz}$$

$$f_1 = 288 \text{ Hz} + 3 \text{ Hz} = 291 \text{ Hz}.$$

The frequency of the string is 291 Hz.

(ii) Required to find the frequency of the loaded string (f_1)

$$f_1 - 288 \text{ Hz} = 2 \text{ Hz}$$

$$f_1 = 2 \text{ Hz} + 288 \text{ Hz} = 290 \text{ Hz}$$

The frequency of the loaded string is 290 Hz.

7. (a) Required to find the frequency of vibration

For the first resonance

$$l_1 = \frac{\lambda}{4}. \quad (\text{End correction is neglected})$$

$$0.12 \text{ m} = \frac{\lambda}{4} \dots \dots \dots \text{eqn(i)}$$

For the second resonance

$$l_2 = \frac{3\lambda}{4}. \quad (\text{End correction is neglected})$$

$$0.37 \text{ m} = \frac{3\lambda}{4} \dots \dots \dots \text{eqn(ii)}$$

Eqn(ii) – eqn (i)

$$0.37 \text{ m} - 0.12 \text{ m} = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$0.25 \text{ m} = \frac{\lambda}{2}$$

$$\lambda = 2 \times 0.25 \text{ m} = 0.50 \text{ m}$$

From the wave equation

$$f = \frac{v}{\lambda} = \frac{340 \text{ m/s}}{0.5 \text{ m}} = 680 \text{ Hz}.$$

Therefore, the frequency of the vibration is 680 Hz.

(b) Required to find the distance between the two men (d)

Distance between men = d

Distance between the second man and cliff = 510 m – d

The time taken by the first sound

$$t_1 = \frac{d}{v}$$

Time taken by the second echo

$$t_2 = \frac{510m + (510m - d)}{v}$$

$$t_2 - t_1 = \frac{(1020m - d)}{v} - \frac{d}{v}$$

$$\Delta t = \frac{1}{v} \times 1020m - 2d$$

$$v\Delta t = 1020m - 2d$$

$$2d = 1020m - 340 \text{ m/s} \times 1.0\text{s} = 680m$$

$$d = \frac{680m}{2} = 340m$$

Therefore, the distance between the men is 340 m.

8. Required to calculate the wavelength of the tuning fork

For the first resonance

$$l_1 = \frac{\lambda}{4}. \quad (\text{End correction is neglected})$$

$$40 \text{ cm} = \frac{\lambda}{4} \dots \dots \dots \text{eqn(i)}$$

For the second resonance

$$l_2 = \frac{3\lambda}{4}. \quad (\text{End correction is neglected})$$

$$85 \text{ cm} = \frac{3\lambda}{4} \dots \dots \dots \text{eqn(ii)}$$

Eqn(ii) – eqn (i)

$$85 \text{ cm} - 40 \text{ cm} = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$45 \text{ cm} = \frac{\lambda}{2}$$

$$\lambda = 2 \times 45 \text{ cm} = 90 \text{ cm}$$

Therefore, the wavelength of the tuning fork is 90 cm.

(b) Required to determine the frequency of the first (f1) and second harmonics (f2).

From

$$f_n = \frac{nv}{2l}$$

$$\text{Where, } v = \sqrt{\frac{T}{\mu}}$$

$$\text{But, } \mu = \frac{\text{mass}}{\text{length}} = \frac{0.002 \text{ kg}}{0.4 \text{ m}} = 0.005 \text{ kg/m}$$

$$v = \sqrt{\frac{100 \text{ N}}{0.005 \text{ kg/m}}} = 141.42 \text{ m/s}$$

$$f_1 = \frac{1 \times 141.42 \text{ m/s}}{2 \times 0.4 \text{ m}} = 176.78/\text{s} = 176.78 \text{ Hz.}$$

$$f_2 = \frac{2 \times 141.42 \text{ m/s}}{2 \times 0.4 \text{ m}} = 353.56/\text{s} = 353.56 \text{ Hz.}$$

Therefore, the frequencies of the first and second harmonics are 176.78 Hz and 353.56 Hz, respectively.

9. (a) The thinnest string will produce the highest frequency. This is because the thinnest string has the lowest linear mass density compared to the thickest string. Since frequency is inversely proportional to the square root of the linear mass density. The thinnest string will give the highest frequency,

$$f \propto \frac{1}{\sqrt{\mu}}$$

(b) Consider a figure below

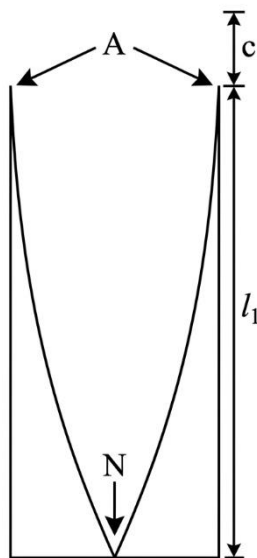


Figure 1.41: The fundamental note (first harmonic)

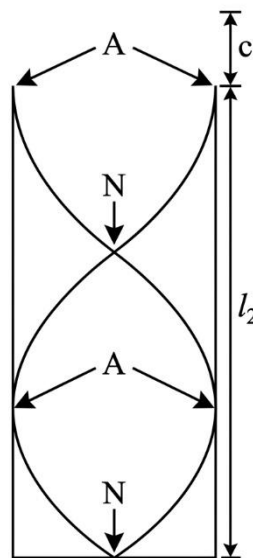


Figure 1.42: Second harmonic (first overtone)

For the fundamental frequency

From wave equation

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$l_1 + c = \frac{\lambda}{4}$$

$$\lambda = 4(l_1 + c)$$

$$f_o = v/4(l_1 + c)$$

For the first overtone

$$l_2 + c = \frac{3\lambda}{4}$$

$$\lambda = 4(l_2 + c)/3$$

$$f_1 = 3v/4(l_2 + c)$$

(c) A big hall or an auditorium is constructed with **acoustic materials** that absorb the sound, these soft materials help to absorb the sound and hence prevent the multiple reflection of sound in a hall. Also, the **use of carpets** can reduce the reflection of sound in a hall.

10. (a) This is because the memory of sound persists in the brain for about 0.1 s, so sound in the small room will take a shorter time than 0.1 s while the original sound is still in the brain.

(b) Required to calculate the tension of the string

From

$$v = \sqrt{\frac{T}{\mu}}$$

$$v^2 = \frac{T}{\mu}$$

$$T = \mu \times v^2$$

$$\text{Where, } \mu = m/l = 0.0004 \text{ kg}/0.4 \text{ m} = 0.001 \text{ kg/m}$$

$$T = 0.001 \text{ kg/m} \times (50 \text{ m/s})^2 = 2.5 \text{ kgm/s}^2 = 2.5 \text{ N}$$

Therefore, the tension of the string is 2.5 N.

(c) This is because light travels faster than sound. The speed of light is about $3 \times 10^8 \text{ m/s}$ while the speed of sound in air is about 340 m/s.

11. (a) $f_1 = 100 \text{ Hz}$

$f_2 = ?$ If $l_2 = 2l_1$

$$\text{Since } f \propto \frac{1}{l}$$

$$fl = k$$

$$f_1 l_1 = f_2 l_2$$

$$100 \text{ Hz} \times l_1 = f_2 \times 2l_1$$

$$f_2 = \frac{100 \text{ Hz}}{2} = 50 \text{ Hz}$$

The frequency will be 50 Hz.

(b) $f_2 = ?$ If $T_2 = 4T_1$

$$f \propto \sqrt{T}$$

$$\frac{f}{\sqrt{T}} = k$$

$$f_1/\sqrt{T_1} = f_2/\sqrt{T_2}$$

$$f_2 = f_1 \times \sqrt{T_2/T_1}$$

$$f_2 = 100 \text{ Hz} \times \sqrt{4T_1/T_1} = 200 \text{ Hz}$$

Therefore, the frequency will be 200 Hz.

12. (a) Required to find the frequencies at each segment

The first segment is formed at a length of 0.50 m

$$l = \frac{\lambda}{2}$$

$$\lambda = 2l = 2 \times 0.50 \text{ m} = 1.0 \text{ m}$$

From wave equation

$$v = f\lambda$$

From the initial condition

$$v = 150/\text{s} \times 2l = 150/\text{s} \times 2 \times 2.0 \text{ m} = 600 \text{ m/s}$$

Now,

$$600 \text{ m/s} = f \times 1.0 \text{ m}$$

$$f = 600/\text{s} = 600 \text{ Hz.}$$

The second segment is at a length of 1.5 m (2.0 m – 0.5 m)

Now

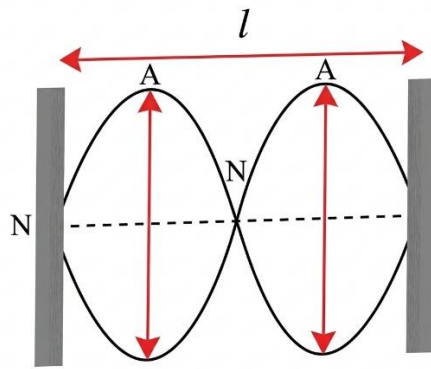
$$v = f\lambda$$

$$600 \text{ m/s} = f \times 2 \times 1.5 \text{ m}$$

$$f = \frac{600 \text{ Hz}}{3} = 200 \text{ Hz.}$$

Therefore, the frequencies produced by the segments are 600 Hz and 200 Hz.

(b) (i) Second harmonic will be formed when a string is set to vibrate by plucking at a point one-quarter of the length from the end as shown in the figure below.



(b) Second harmonic (First overtone)

(ii) From wave equation

$$v = f\lambda$$

$$f = \frac{v}{\lambda}$$

$$l = \lambda \text{ (for second harmonic)}$$

$$f = \frac{v}{l}$$

13. (a) Required to find the location of Y

$$\text{Since } f \propto \frac{1}{l}$$

$$fl = k$$

$$f_1 l_1 = f_2 l_2$$

$$320 \text{ Hz} \times 0.80 \text{ m} = 256 \text{ Hz} \times l_2$$

$$l_2 = \frac{256 \text{ m}}{256} = 1 \text{ m.}$$

Y must be placed at a distance of 0.2 m further away (1 m – 0.8 m).

(b) Required to calculate the distance between the men

From the formula

$$\text{Speed of sound (v)} = \frac{\text{Distance covered (d)}}{\text{Time taken (t)}}$$

$$t = \frac{d}{v}$$

When sound travels through air when

$$t_1 = d/v_a$$

When sound travels through the metal

$$t_2 = d/v_m$$

Since sound travels faster in metal (solid) than air (gas)

$$t_1 - t_2 = \Delta t = d/v_a - d/v_m = d \times (1/v_a - 1/v_m)$$

$$\frac{\Delta t}{d} = 1/v_a - 1/v_m$$

$$\frac{\Delta t}{d} = (v_m - v_a)/(v_a \times v_m)$$

$$d = \Delta t \times (v_a \times v_m)/(v_m - v_a)$$

$$d = 0.5 \text{ s} \times \frac{(330 \frac{m}{s} \times 5280 \frac{m}{s})}{(5280 \frac{m}{s} - 330 \text{ m/s})}$$

$$d = 176 \text{ m.}$$

Therefore, the distance between the two men is 176 m.

14. (a) Required to determine the frequency of the fundamental note

From

$$f = \frac{1}{2l} \times \sqrt{\frac{T}{\mu}}$$

$$\mu = \frac{m}{l}$$

$$f = \frac{1}{2l} \times \sqrt{\frac{Tl}{m}}$$

$$f = \frac{1}{2 \times 0.60 \text{ m}} \times \sqrt{\frac{960 \text{ N} \times 0.60 \text{ m}}{0.01 \text{ kg}}}$$

$$f = 200 \text{ Hz}$$

Therefore, the frequency of the fundamental note is 200 Hz.

(b) (i) Region E represents ultraviolet which promotes the production of vitamin D.

(ii) Region A represents radio waves which is the radiation used in RADAR system.

(iii) Region G represents gamma rays which produced in nuclear reactor.

15. (a) Required to find the beat frequency

$$f \propto \sqrt{T}$$

$$\frac{f}{\sqrt{T}} = k$$

$$f_1/\sqrt{T_1} = f_2/\sqrt{T_2}$$

$$f_2 = f_1 \times \sqrt{T_2/T_1}$$

$$\text{Where, } f_1 = 384 \text{ Hz, } T_2 = T_1 + \frac{2}{100} \times T_1 = 1.02T_1$$

$$f_2 = 384 \text{ Hz} \times \sqrt{1.02T_1/T_1} = 387.82 \text{ Hz}$$

$$\text{Beat frequency, } f_b = f_2 - f_1 = 387.82 \text{ Hz} - 384 \text{ Hz} = 3.82 \text{ Hz}$$

Therefore, the beat frequency is 3.82 Hz.

(b) (i) Required to determine the frequency of the wave

$$\text{Frequency} = \frac{1}{\text{Period}}$$

Period (T) = time taken to complete one cycle = $0.005 \text{ s} - 0.001 \text{ s} = 0.004 \text{ s}$

$$\text{Frequency} = \frac{1}{0.004 \text{ s}} = 250 \text{ Hz}$$

Therefore, the frequency of the wave is 250 Hz.

(ii) Required to determine the velocity of the wave

From wave equation

$$v = f\lambda$$

Wavelength = distance between two successive crests or troughs

$$\lambda = 10 \text{ m} - 2 \text{ m} = 8 \text{ m}$$

$$v = 250/\text{s} \times 8 \text{ m} = 2000 \text{ m/s}$$

therefore, the velocity of the wave is 2000 m/s

16. (a) Required the frequency of the wave

From the wave equation

$$v = f\lambda$$

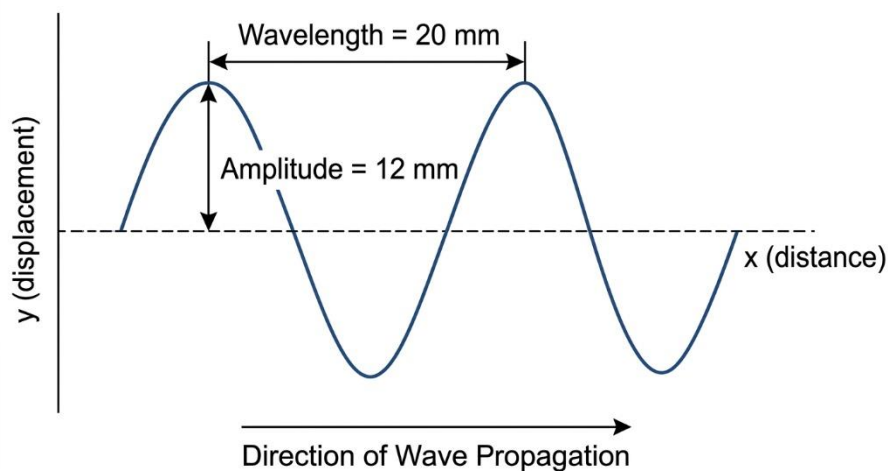
where, $v = 16 \text{ cm/s}$ and $\lambda = 20 \text{ mm} = 2.0 \text{ cm}$

$$16 \text{ cm/s} = f \times 2.0 \text{ cm}$$

$$\frac{16 \text{ cm/s}}{2.0 \text{ cm}} = 8/\text{s} = 8 \text{ Hz} = f$$

Therefore, the frequency of the waves is 8 Hz.

(b) A diagram that shows the side-view of two complete waves.



17. (a) Required to find speed of light in glass

From

$$\eta = \frac{c}{v}$$

where, η is the refractive index of glass, c is the speed of light in vacuum and v is the speed of light in glass

$$v = \frac{c}{\eta} = 3.00 \times 10^8 \text{ ms}^{-1} / 1.5 = 2.00 \times 10^8 \text{ m/s}$$

Therefore, the speed of light in glass is $2.00 \times 10^8 \text{ m/s}$.

(b) Required to find the wavelength of light in glass

From

$$\eta = \frac{c}{v}$$

But, speed of wave = $f\lambda$, where frequency of wave is always constant

$$\eta = f\lambda / f\lambda_1 = \lambda / \lambda_1$$

where, λ and λ_1 are wavelengths of wave in vacuum and in glass, respectively

$$\lambda_1 = \frac{\lambda}{\eta} = 6.0 \times 10^{-7} \text{ m} / 1.5 = 4.0 \times 10^{-7} \text{ m}$$

Therefore, the wavelength of light in glass is $4.0 \times 10^{-7} \text{ m}$.

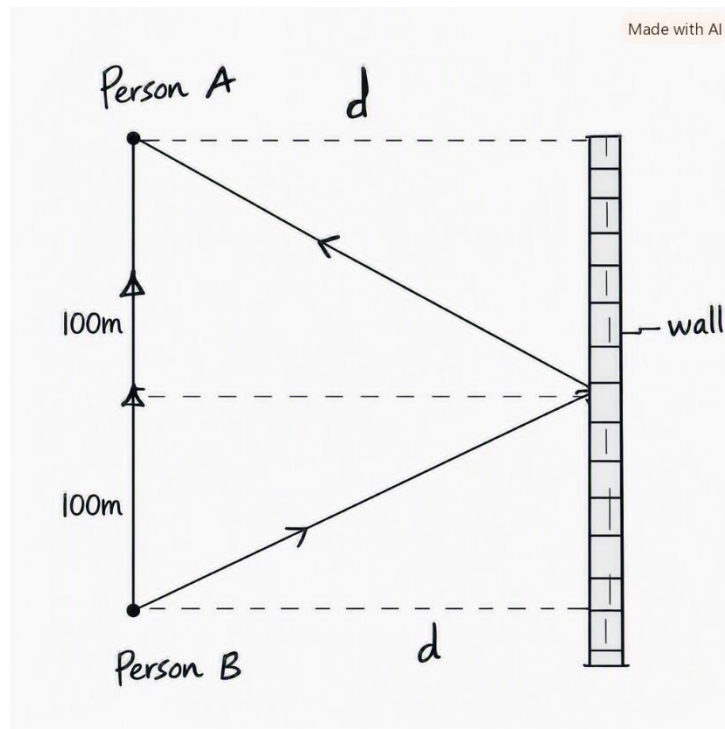
18. (a) Amplitude is the maximum displacement which is 8 mm.

(b) Period of the wave = time taken to complete one cycle = 0.2 s

(c) The frequency of the wave = $\frac{1}{\text{period}} = \frac{1}{0.2 \text{ s}} = 5/\text{s} = 5 \text{ Hz}$.

(d) Velocity of the wave = $f\lambda = 5/\text{s} \times 18 \text{ mm} = 90 \text{ mm/s} = 0.09 \text{ m/s}$.

19. (a) Consider a figure below showing the positions of the persons



(b) Required to find the velocity of sound in air

From

$$v = \frac{\text{Distance between two persons}}{\text{time taken by the first sound}}$$
$$v = \frac{200 \text{ m}}{0.60 \text{ s}} = 333.33 \text{ m/s}$$

Therefore, the velocity of sound in air is 333.33 m/s

(c) Required to find the perpendicular distance of the persons from the wall

The second sound heard by the person A after person B fired the pistol is an echo

$$\text{Velocity of sound in air under echo case} = \frac{2 \times \text{distance travelled to hit the wall}}{\text{time taken}}$$

$$\text{Time taken} = 0.6 \text{ s} + 0.25 \text{ s} = 0.85 \text{ s}$$

$$\text{Distance travelled to hit the wall} = \frac{vt}{2} = \frac{333.33 \frac{\text{m}}{\text{s}} \times 0.85 \text{ s}}{2} = 141.67 \text{ m}$$

Consider one of the right-angled triangles in the drawn diagram above

From Pythagoras theorem

$$d^2 + 100^2 \text{ m}^2 = 141.67^2 \text{ m}^2$$

$$d = \text{m} \sqrt{(141.67^2 - 100^2)} = 100.35 \text{ m}$$

Therefore, the perpendicular distance of the persons from the wall is 100.35 m.

20. (a) Required to find the ratio of velocities

From

$$v = (331.4 + 0.6\theta) \text{ m/s}$$

$$\text{when } \theta = 42^\circ\text{C}$$

$$v_1 = (331.4 + 0.6 \times 42) \text{ m/s} = 356.6 \text{ m/s}$$

$$\text{when } \theta = 60^\circ\text{C}$$

$$v_2 = (331.4 + 0.6 \times 60) \text{ m/s} = 367.4 \text{ m/s}$$

$$v_1/v_2 = \frac{356.6 \text{ m/s}}{367.4 \text{ m/s}} = \frac{1783}{1837}$$

$$v_1 : v_2 = 1783 : 1837$$

Therefore, the ratio of velocities is 1783:1837

(b) (i) When a tuning fork is sounded at the open end of a tube with the other end closed, the air in the tube resonates (vibrates freely) at a certain length of the tube. The resonance is observed as a loud sound produced in the tube when the proper length of air column is obtained. The first resonance occurs when air vibrates at its fundamental note or first harmonic.

(ii) Required to find the frequency of the vibration

For the first resonance

$$l_1 = \frac{\lambda}{4}. \quad (\text{End correction is neglected})$$

$$0.22 \text{ m} = \frac{\lambda}{4} \dots \dots \dots \text{eqn(i)}$$

For the second resonance

$$l_2 = \frac{3\lambda}{4}. \quad (\text{End correction is neglected})$$

$$0.47 \text{ m} = \frac{3\lambda}{4} \dots \dots \dots \text{eqn(ii)}$$

Eqn(ii) – eqn (i)

$$0.47 \text{ m} - 0.22 \text{ m} = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$0.25 \text{ m} = \frac{\lambda}{2}$$

$$\lambda = 2 \times 0.25 \text{ m} = 0.5 \text{ m}$$

From wave equation

$$v = f\lambda$$

$$f = \frac{v}{\lambda} = \frac{330 \text{ m/s}}{0.5 \text{ m}} = 660 \text{ Hz}$$

Therefore, the frequency of vibration is 660 Hz.

21. (a) Required to find how long will take for the beat to reach the audience member

From

$$v = \frac{\text{distance between beat source and the audience}}{\text{time taken}}$$

$$t = \frac{d}{v}$$

$$\text{But, } v = (331.4 + 0.6\theta) \text{ m/s}$$

$$\text{when } \theta = 21^\circ\text{C}$$

$$v = (331.4 + 0.6 \times 21) \text{ m/s} = 344 \text{ m/s}$$

$$t = \frac{100 \text{ m}}{344 \text{ m/s}}$$

$$t \approx 0.29 \text{ s}$$

Therefore, it will take about 0.29 s for the beat to reach the audience member.

(b) Required to find the depth of the sea

From

$$v = \frac{2d}{t}$$

where, v is the speed of sound in water, d is the depth of the ocean and t is time taken for the sound to return.

$$d = \frac{vt}{2} = \frac{1500 \frac{m}{s} \times 1 s}{2} = 750 \text{ m}$$

Therefore, the depth of the ocean is 750 m.

22. Required to find the distance between the cliffs

From

$$v = \frac{2 \times \text{distance between an observer}}{t}$$

From near cliff

$$d_1 = \frac{vt}{2}$$

$$d_1 = \frac{330 \frac{m}{s} \times 2 s}{2} = 330 \text{ m}$$

From far cliff

$$d_2 = \frac{330 \frac{m}{s} \times 2.5 s}{2} = 412.5 \text{ m}$$

$$\text{Distance between two cliffs} = d_1 + d_2 = 330 \text{ m} + 412.5 = 742.5 \text{ m}$$

Therefore, the distance cliffs is 742.5 m.

23. (a) This is because both radio waves and light waves are electromagnetic waves while sound waves are mechanical waves. Mechanical waves require material medium in order to propagate while electromagnetic waves do not require material medium to propagate.

(b) Required to determine the wavelength and end correction.

From

For the first resonance

$$l_1 + c = \frac{\lambda}{4}$$

$$23 \text{ cm} + c = \frac{\lambda}{4} \dots \dots \dots \text{eqn(i)}$$

For the second resonance

$$l_2 + c = \frac{3\lambda}{4}$$

$$73 \text{ cm} + c = \frac{3\lambda}{4} \dots \dots \dots \text{eqn(ii)}$$

$$\text{Eqn(ii)} - \text{eqn (i)}$$

$$73 \text{ cm} - 23 \text{ cm} = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$50 \text{ cm} = \frac{\lambda}{2}$$

$$\lambda = 2 \times 50 \text{ cm} = 100 \text{ cm}$$

From the first equation

$$23 \text{ cm} + c = \frac{100 \text{ cm}}{4}$$

$$c = 25 \text{ cm} - 23 \text{ cm} = 2 \text{ cm}.$$

Therefore, the wavelength of the sound and end correction are 100 cm and 2 cm, respectively.

(c) This is because the temperature change affects the length of strings due to thermal expansion. Length of string increases when temperature increases and decreases when temperature decreases. This affects the frequency produced when a string instrument is played.

$$\text{Frequency} \propto \frac{1}{l}$$

Thermal expansion also affects the tension in the string. When temperature increases, the string becomes slack and hence the tension is lowered and vice versa is true.

$$\text{Frequency} \propto \sqrt{T}$$

24. (a) You should aim below it, because the fish appears to be closer due to refraction of light.

(b) Both speed and wavelength change when wave move from one medium to another. For example, speed and wavelength increase when wave such as light move from denser medium (glass) to a less dense medium (air).

25. (a) Required to determine the distance between adjacent wave crests

From

$$\text{Distance between adjacent wave crests} = \text{wavelength} = \frac{v}{f}$$

$$\lambda = \frac{5 \text{ m/s}}{2/\text{s}} = 2.5 \text{ m}$$

Therefore, the distance between adjacent wave crests is 2.5 m.

(b) Required to find how long it takes for a girl to hear an echo.

From

$$v = \frac{2d}{t}$$

$$t = \frac{2d}{v} = \frac{2 \times 250 \text{ m}}{340 \text{ m/s}}$$

$$t \approx 1.47 \text{ s}$$

Therefore, it will take about 1.47 s for a girl to hear the echo.

26. Required to find the speed of the wave in the second medium

From

$$v = f\lambda$$

Since frequency is always constant

in the first medium

$$v_1 = f\lambda_1$$

in the second medium

$$v_2 = f\lambda_2$$

$$v_2/v_1 = f\lambda_2/f\lambda_1$$

$$v_2 = v_1 \times (\lambda_2/\lambda_1)$$

where, $v_1 = 4.4 \text{ m/s}$, $\lambda_1 = 2 \text{ m}$ and $\lambda_2 = 3 \text{ m}$

$$v_2 = 4.4 \text{ m/s} \times \frac{3 \text{ m}}{2 \text{ m}} = 6.6 \text{ m/s}$$

Therefore, the speed of the wave in the second medium is 6.6 m/s.

27. Required to calculate the wavelength of the sinusoidal wave

From wave equation

$$v = f\lambda$$

$$\lambda = \frac{v}{f}$$

$$\text{But, } f = \frac{\text{number of vibrations}}{\text{time taken}} = \frac{40}{30 \text{ s}}$$

$$f = \frac{4}{3} \text{ Hz}$$

$$\text{Also, } v = \frac{\text{Distance covered by the wave}}{\text{time taken}} = \frac{425 \text{ cm}}{10 \text{ s}}$$

$$v = 42.5 \text{ cm/s}$$

$$\lambda = \frac{42.5 \text{ cm/s}}{\frac{4}{3}/\text{s}} = 31.875 \text{ cm}$$

Therefore, the wavelength of the sinusoidal wave is 31.875 cm.

28. Data given

Velocity of light in water, $v_w = 2.2 \times 10^8 \text{ m/s}$

Velocity of light in glass, $v_g = 2.0 \times 10^8 \text{ m/s}$

Required to calculate

(a) The relative refractive index when light passes from water to glass.

From

$${}_w\eta_g = v_w/v_g$$

$${}_w\eta_g = 2.2 \times 10^8 \text{ ms}^{-1} / 2.0 \times 10^8 \text{ ms}^{-1}$$

$${}_w\eta_g = 1.1$$

Therefore, the relative refractive index when light passes from water to glass is 1.1.

- (b) The angle of incidence in the water which would produce an angle of refraction of 30° in the glass

From Snell's law

$${}_w\eta_g = \frac{\sin i}{\sin r}$$

$$i = \sin^{-1}({}_w\eta_g \times \sin r)$$

$$i = \sin^{-1}(1.1 \times \sin 30^\circ)$$

$$i = 33.37^\circ$$

Therefore, the angle of incidence in the water is 33.37° .

29. Required to calculate the frequency of the vibrator

From the wave equation

$$v = f\lambda$$

$$v = \frac{\text{Distance covered by the wave}}{\text{Time taken}}$$

$$v = \frac{33 \text{ cm}}{1.5 \text{ s}} = 22 \text{ cm/s}$$

wavelength, λ = distance between two successive crests = 4.0 cm

$$22 \text{ cm/s} = f \times 4.0 \text{ cm}$$

$$f = \frac{22 \text{ cm/s}}{4.0 \text{ cm}} = 5.5/\text{s} = 5.5 \text{ Hz}$$

Therefore, the frequency of the vibrator is 5.5 Hz.

30. Required to calculate

- (a) The wavelength of radio waves

From the wave equation

$$v = f\lambda$$

Where, $v = 3 \times 10^8 \text{ m/s}$ and $f = 150 \text{ MHz} = 1.5 \times 10^8 \text{ Hz}$.

$$3 \times 10^8 \text{ m/s} = 1.5 \times 10^8/\text{s} \times \lambda$$

$$\lambda = 2 \text{ m.}$$

Therefore, the wavelength of the radio wave is 2 m.

- (b) The velocity of light in glass of refractive index 1.5.

From

$$\eta = \frac{\text{velocity of light in air}}{\text{velocity of light in glass}}$$

$$v_g = v_a / \eta = 3.0 \times 10^8 \text{ ms}^{-1} / 1.5 = 2.0 \times 10^8 \text{ m/s}$$

Therefore, the velocity of light in glass is $2.0 \times 10^8 \text{ m/s}$.

31. (a) Required to find the speed of the wave

From the wave equation

$$v = f\lambda$$

$$f = \frac{\text{number of waves}}{\text{time taken}}$$

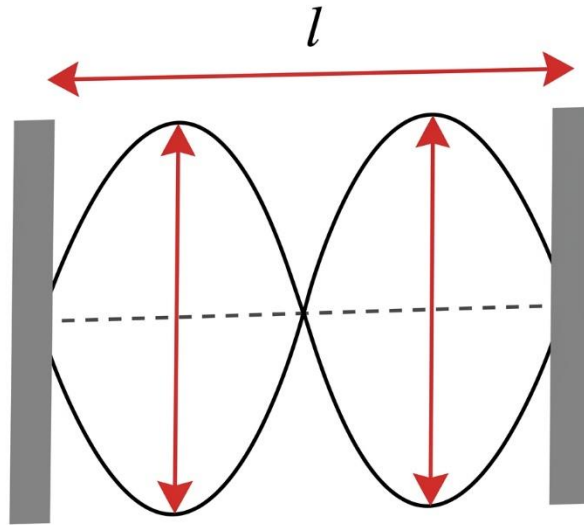
$$f = \frac{3}{1 \text{ s}} = 3/\text{s}$$

$$\lambda = \text{Distance between two successive crests} = 2 \text{ m}$$

$$v = 3/\text{s} \times 2 \text{ m} = 6 \text{ m/s}$$

Therefore, the speed of the wave is 6 m/s .

(b) The ceilings and walls are covered with perforated boards in order to prevent or reduce both echo and reverberation.



32. (a)

(b) Wavelength of the wave = $l = 1.2 \text{ m}$

33. Required to determine

(a) The fundamental frequency

From

$$f = \frac{1}{2l} \times \sqrt{\frac{T}{\mu}}$$

$$\text{where, } \mu = \frac{\text{mass of the wire}}{\text{Length of the wire}}$$

$$\mu = \frac{1.2 \text{ g}}{20 \text{ cm}} = \frac{0.0012 \text{ kg}}{0.2 \text{ m}} = 0.006 \text{ kg/m}$$

$$f = \frac{1}{2 \times 0.2 \text{ m}} \times \sqrt{\frac{120 \text{ N}}{0.006 \text{ kg/m}}}$$

$$f = 353.55 \text{ Hz}$$

Therefore, the fundamental frequency is 353.55 Hz.

(b) The frequency of the third harmonic

From

$$f_3 = 3 \times f_0$$

$$f_3 = 3 \times 353.55 \text{ Hz} = 1060.65 \text{ Hz}.$$

Therefore, the frequency of the third harmonic is 1060.65 Hz.

34. Required to determine

(a) The speed of sound in air

From the wave equation

$$v = f\lambda$$

For the first resonance

$$l_1 + c = \frac{\lambda}{4}.$$

$$31 \text{ cm} + c = \frac{\lambda}{4} \dots \dots \dots \text{eqn(i)}$$

For the second resonance

$$l_2 + c = \frac{3\lambda}{4}.$$

$$99.8 \text{ cm} + c = \frac{3\lambda}{4} \dots \dots \dots \text{eqn(ii)}$$

Eqn(ii) – eqn (i)

$$99.8 \text{ cm} - 31 \text{ cm} = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$68.8 \text{ cm} = \frac{\lambda}{2}$$

$$\lambda = 2 \times 68.8 \text{ cm} = 137.6 \text{ cm}$$

$$v = 250/\text{s} \times 137.6 \text{ cm} = 34400 \text{ cm/s} = 344 \text{ m/s}$$

Therefore, the speed of the sound in air is 344 m/s.

(b) The end correction

From the first equation

$$31 \text{ cm} + c = \frac{\lambda}{4}$$

$$c = \frac{137.6 \text{ cm}}{4} - 31 \text{ cm}$$

$$c = 3.4 \text{ cm}$$

Therefore, the end correction is 3.4 cm.

35. Required to find the frequency of vibration

From the wave equation

$$v = f\lambda$$

For the first resonance

$$l_1 = \frac{\lambda}{4}. \quad (\text{End correction is neglected})$$

$$0.12 \text{ m} = \frac{\lambda}{4} \dots \dots \dots \text{eqn(i)}$$

For the second resonance

$$l_2 = \frac{3\lambda}{4}. \quad (\text{End correction is neglected})$$

$$0.37 \text{ m} = \frac{3\lambda}{4} \dots \dots \dots \text{eqn(ii)}$$

Eqn(ii) – eqn (i)

$$0.37 \text{ m} - 0.12 \text{ m} = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$0.25 \text{ m} = \frac{\lambda}{2}$$

$$\lambda = 2 \times 0.25 \text{ m} = 0.50 \text{ m}$$

Now

$$340 \text{ m/s} = f \times 0.50 \text{ m}$$

$$680 \text{ Hz} = f$$

Therefore, the frequency of vibration is 680 Hz.

36. Required to find the frequency of the tuning fork

From the wave equation

$$v = f\lambda$$

$$l = \frac{\lambda}{4} \text{ (First resonance)}$$

$$\lambda = 4l = 4 \times 26.25 \text{ cm} = 105 \text{ cm}$$

Now

$$33600 \text{ cm/s} = f \times 105 \text{ cm}$$

$$f = \frac{33600 \text{ cm/s}}{105 \text{ cm}} = 320/\text{s} = 320 \text{ Hz.}$$

Therefore, the frequency of the tuning fork is 320 Hz.

37. Required to find how far is the storm cloud

From

$$t = \frac{d}{v}$$

For light case

Speed of light in air, $v_1 = 3.0 \times 10^8 \text{ m/s}$

$$t_1 = d/3.0 \times 10^8 \text{ ms}^{-1}$$

For sound case

$$v = (331.4 + 0.6\theta)$$

$$v_2 = (331.4 + 0.6 \times 28) \text{ m/s} = 348.2 \text{ m/s}$$

$$t_2 = d/348.2 \text{ ms}^{-1}$$

$$\Delta t = t_2 - t_1$$

$$\Delta t = d/v_2 - d/v_1$$

$$\frac{\Delta t}{d} = 1/v_2 - 1/v_1$$

$$\frac{\Delta t}{d} = (v_1 - v_2)/(v_2 \times v_1)$$

$$d = \Delta t \times (v_2 \times v_1)/(v_1 - v_2)$$

$$d = 7 \text{ s} \times (348.2 \text{ m/s} \times 3.0 \times 10^8 \text{ m/s})/(3.0 \times 10^8 \text{ m/s} - 348.2 \text{ m/s})$$

$$d = 2437.40 \text{ m}$$

Therefore, the storm cloud is at a distance of 2437.40 m.

38. Required to find the distance between sound source and echo

From

$$v = \frac{2d}{t}$$

$$d = \frac{vt}{2}$$

$$v = (331.4 + 0.6\theta)$$

$$v = (331.4 + 0.6 \times 22) \text{ m/s} = 344.6 \text{ m/s}$$

$$d = \frac{344.6 \frac{\text{m}}{\text{s}} \times 8 \text{ s}}{2} = 1378.4 \text{ m}$$

Therefore, the cliff is 1378.4 m away from the cliff.

39. Required to find the length of string B

For a string B to resonate with string A

Frequency of string A = frequency of string B

$$\text{But, } f = \frac{1}{2l} \times \sqrt{\frac{T}{\mu}}$$

$$\frac{1}{2 \times 2 \text{ m}} \times \sqrt{\frac{T}{9 \text{ g/cm}}} = \frac{1}{2l} \times \sqrt{\frac{T}{18 \text{ g/cm}}}$$

$$l = 2 \text{ m} \times \sqrt{\frac{9 \text{ g/cm}}{18 \text{ g/cm}}} = 1.41 \text{ m}$$

Therefore, the length of string B is 1.41 m.

40. Required to find the length when the second resonance occurs

For first resonance

$$l_1 = \frac{\lambda}{4}$$

$$\lambda = 4l_1 = 4 \times 9.0 \text{ cm} = 36.0 \text{ cm}$$

for the second resonance

$$l_2 = \frac{3\lambda}{4}$$

$$l_2 = \frac{3 \times 36.0 \text{ cm}}{4} = 27.0 \text{ cm}$$

$$l_2 - l_1 = 27.0 \text{ cm} - 9.0 \text{ cm} = 18.0 \text{ cm}$$

Therefore, the tube must be raised by 18.0 cm.

41. This is because a bridge may start resonate and vibrate at high amplitude and hence breaks down.

42. Required to find

(a) Length

$$\text{Since } f \propto \frac{1}{l}$$

$$f_1 l_1 = f_2 l_2$$

$$250 \text{ Hz} \times 87 \text{ cm} = 375 \text{ Hz} \times l_2$$

$$l_2 = 58 \text{ cm}$$

(b) Tension

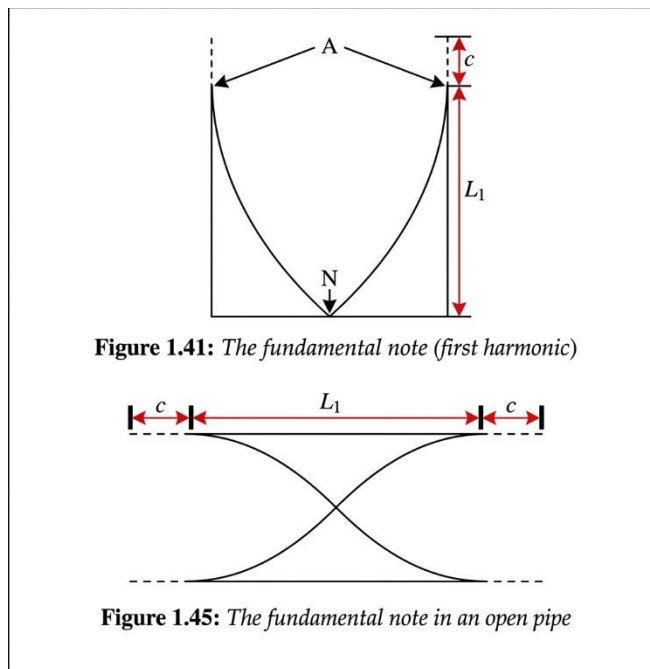
$$\text{Since } f \propto \sqrt{T}$$

$$f_1 / \sqrt{T_1} = f_2 / \sqrt{T_2}$$

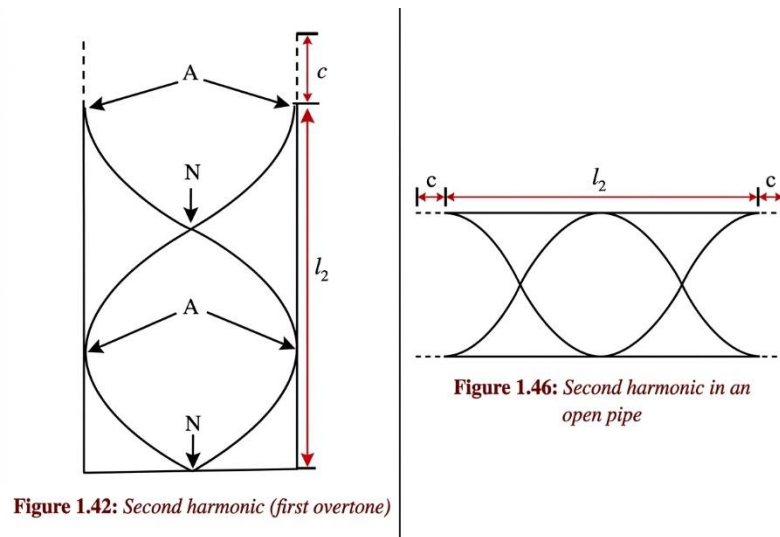
$$T_2 = T_1 \times (f_2 / f_1)^2$$

$$T_2 = 10 \text{ N} \times \left(\frac{375 \text{ Hz}}{250 \text{ Hz}}\right)^2 = 22.5 \text{ N}.$$

43. (a) Figure below shows the mode of vibration for fundamental note in organ pipe A and B



44. (b) Figure below shows the mode of vibration for first overtones in organ pipe A and B



45. (a) (i) Light source, oscillating paddle, power source and water tank
 (ii) The role of stroboscope is to slow down the motion of waves so as to make possible for an observer to see the behaviours of waves.
- (i) This is because the frequency produced by each string is different. Thick wire produces low frequency while thin wire produces high frequency.
- (ii) By altering the length of the string where the shortest length produces highest frequency while longer wire produces lowest frequency.

$$f \propto \frac{1}{l}$$

46. (a) This is because each instrument produces different overtones.

(b) This is because flute is longer than a piccolo. Frequency is inversely proportional to the length (length of air column).